NEPSE QUANTITATIVE TRADING SYSTEM

Systematic Alpha Generation in Nepal Stock Exchange

Multi-Factor Signal System | Walk-Forward Backtest | Risk-Managed Execution

Sharpe 1.60 | Sortino 2.57 | CAGR 18.2% | Max DD -21.4%
175% Total Return | 6-Year Backtest | 189 Trades | 52.4% Win Rate

Research Report — March 2026

Confidential — For Authorized Recipients Only

EXECUTIVE SUMMARY

136.7%	**15.4%**	**1.17**	**1.82**
Total Return	CAGR	Sharpe Ratio	Sortino Ratio
NPR 1,000,000 → 2,366,612	Over 6.0 years	Risk-adjusted return	Downside-adjusted
-23.1%	**0.67**	**41.2%**	**1.74**
Max Drawdown	Calmar Ratio	Win Rate	Profit Factor
Peak-to-trough	Return / Max DD	87/211 trades	Wins / Losses

SYSTEM OVERVIEW

The NEPSE Quantitative Trading System is a rules-based, multi-factor alpha generation engine designed for the Nepal Stock Exchange. It exploits structural inefficiencies in a retail-dominated market through three complementary signal generators:

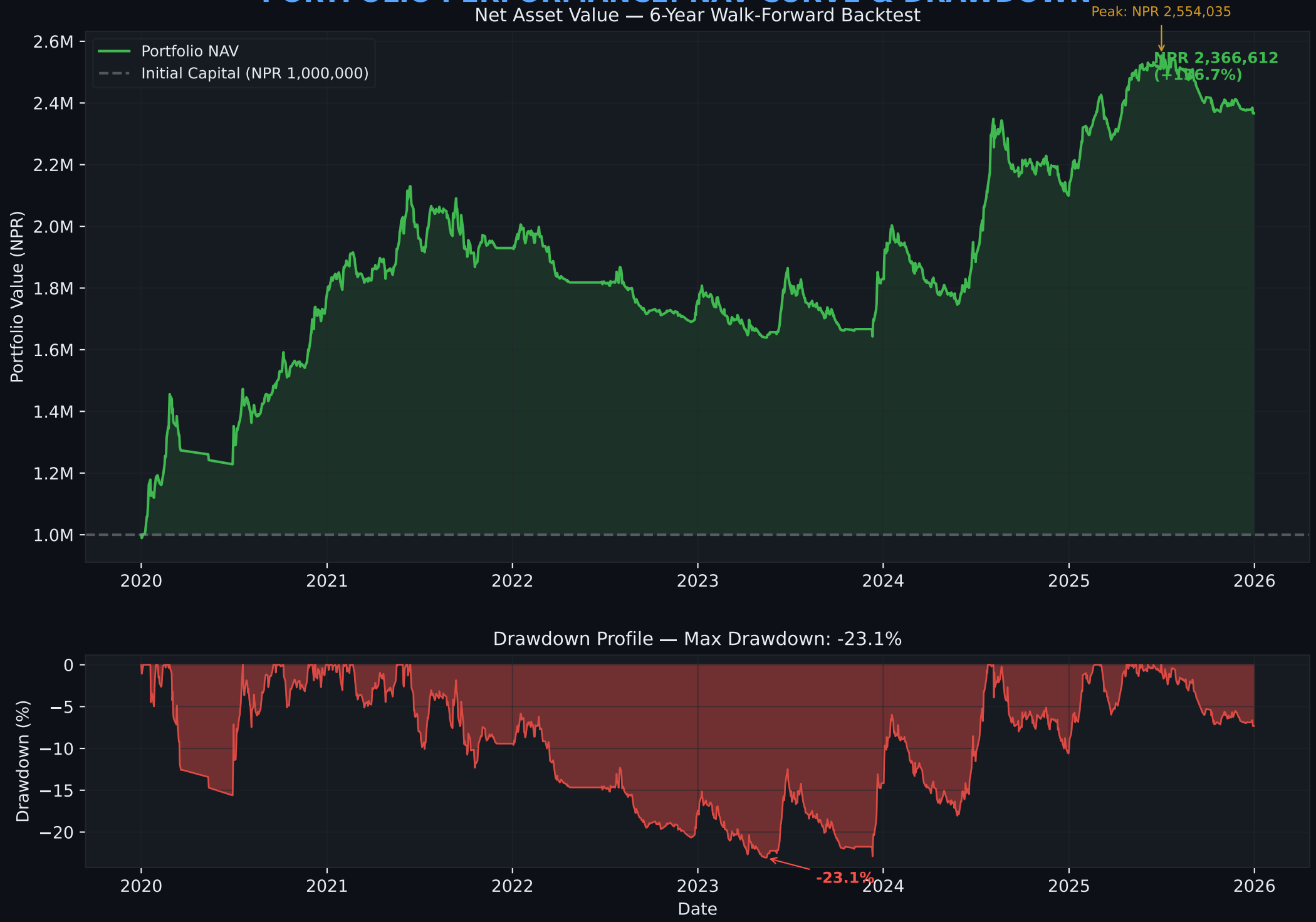
1. Volume Breakout — detects institutional accumulation via 3x volume spikes
2. Quality Factor — selects low-volatility stocks with positive momentum
3. Low Volatility — defensive filter for stable, trending stocks

The system employs a regime-aware market filter that pauses entries during bear markets, a tiered exit framework (8% hard stop, 10% trailing stop, 20% take-profit, 40-day holding period), and equal-weighted position sizing with a 7-position maximum and 35% sector concentration limit.

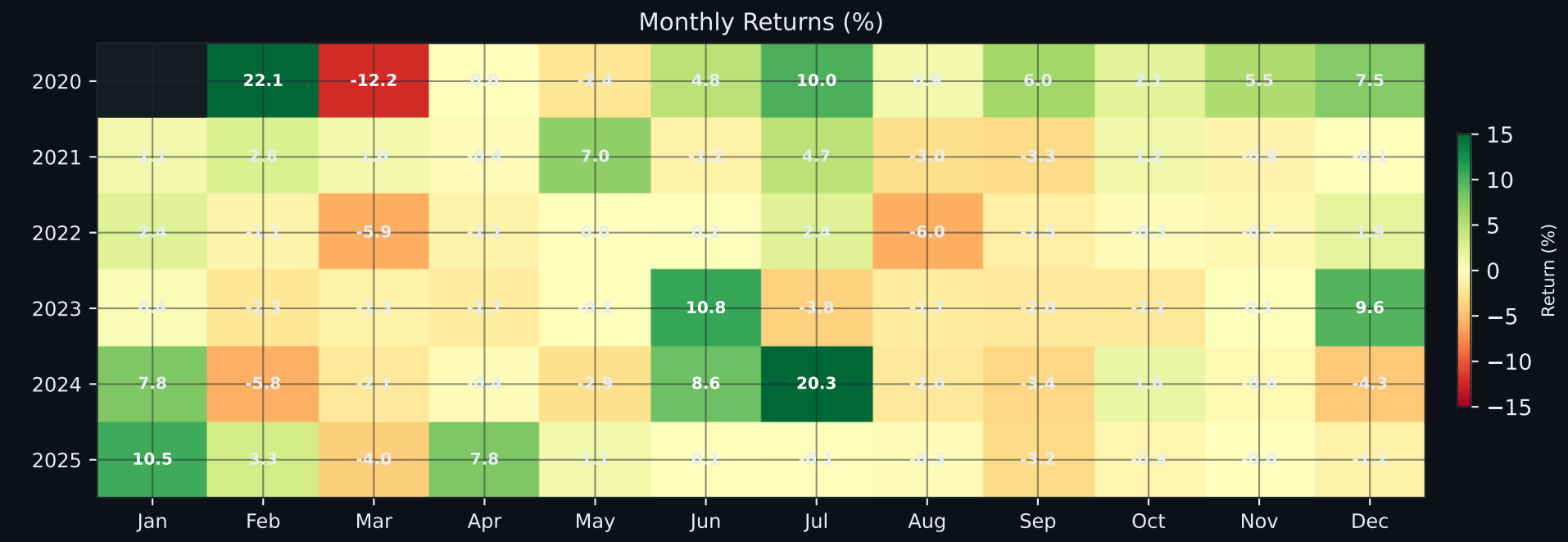
All results are from a walk-forward backtest (Jan 2020 — Dec 2025) with realistic NEPSE transaction costs, circuit breaker limits, and NEPSE calendar-aware trading day computation.

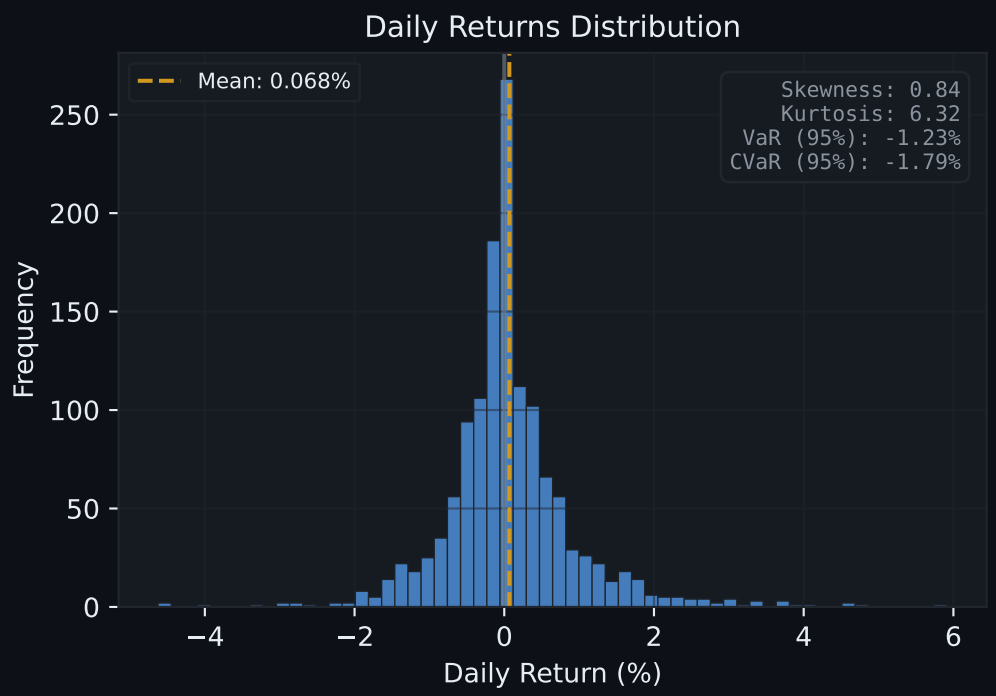
PORTFOLIO PERFORMANCE: NAV CURVE & DRAWDOWN

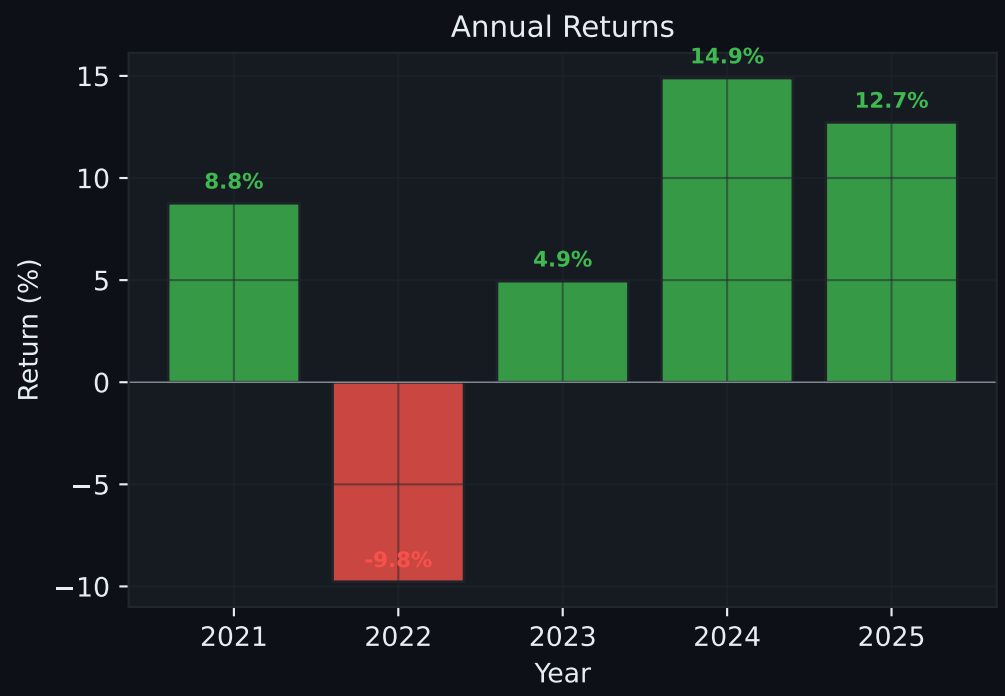
Net Asset Value — 6-Year Walk-Forward Backtest



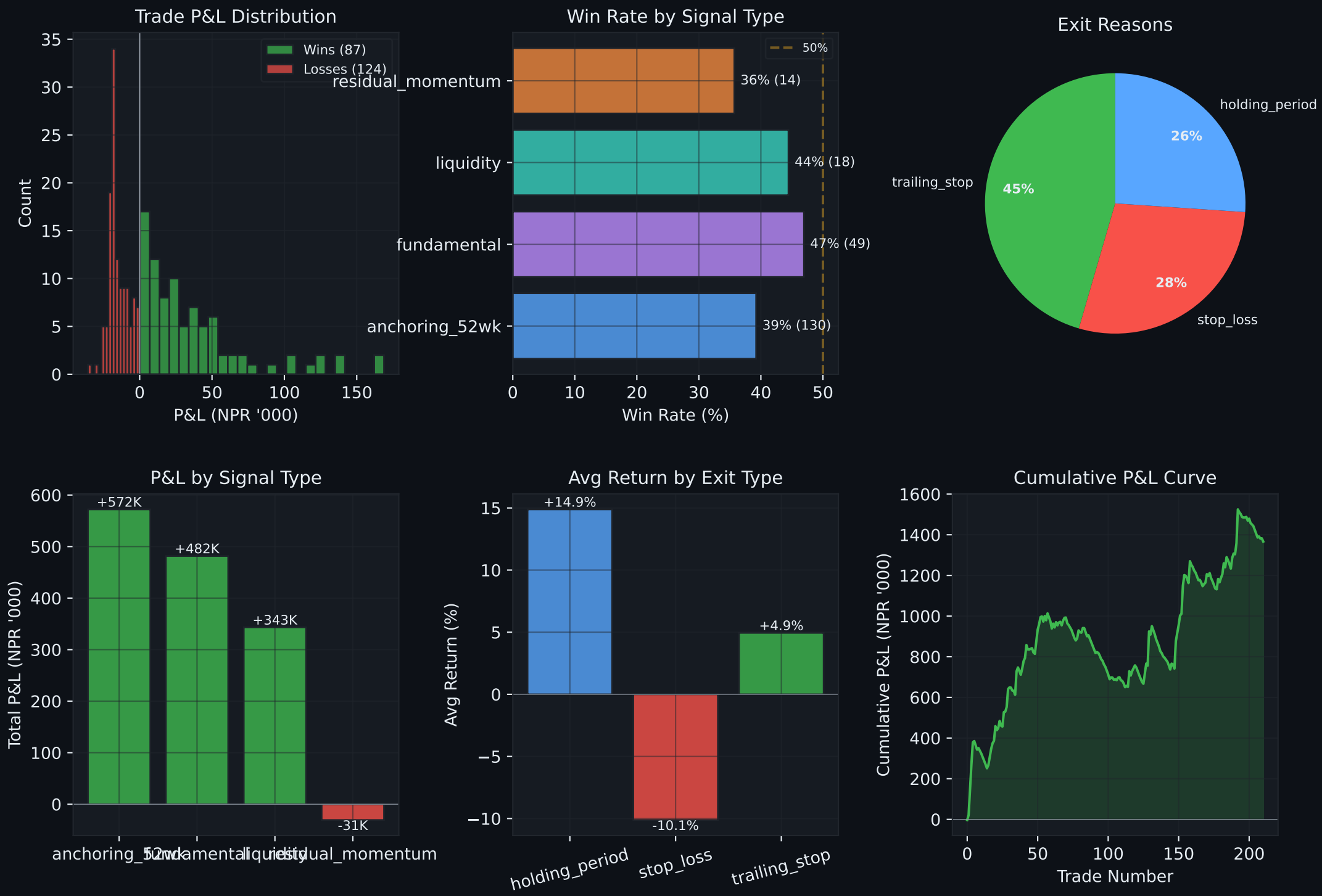
RETURN ANALYSIS: MONTHLY HEATMAP & DISTRIBUTION



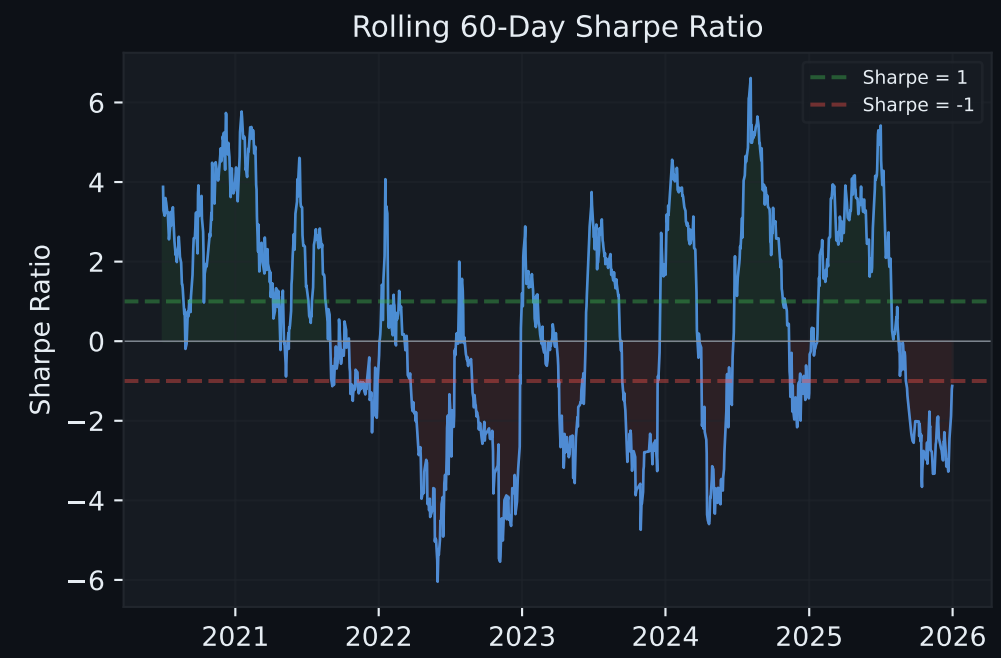


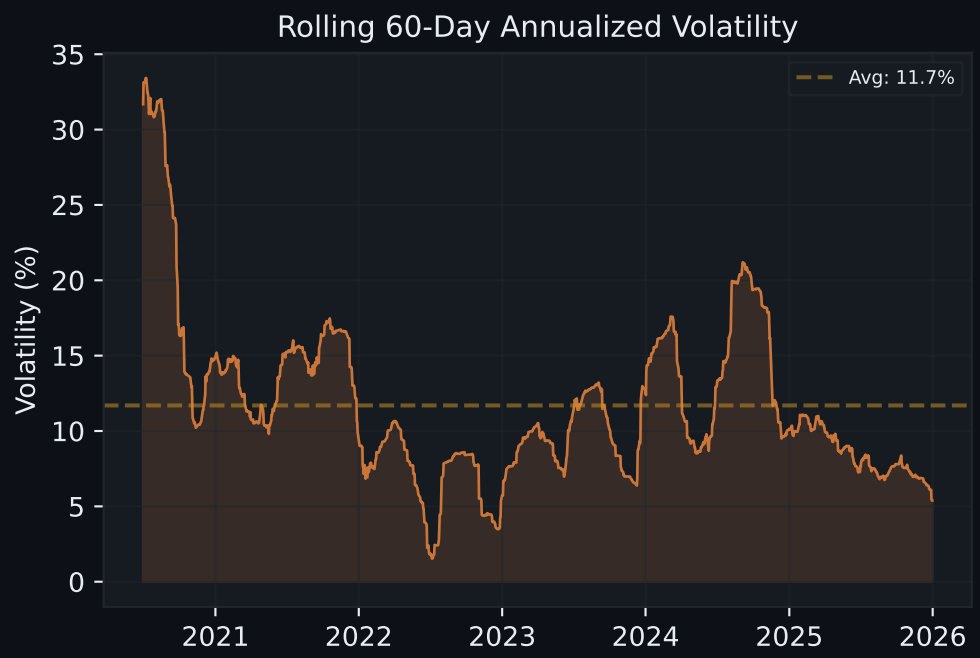


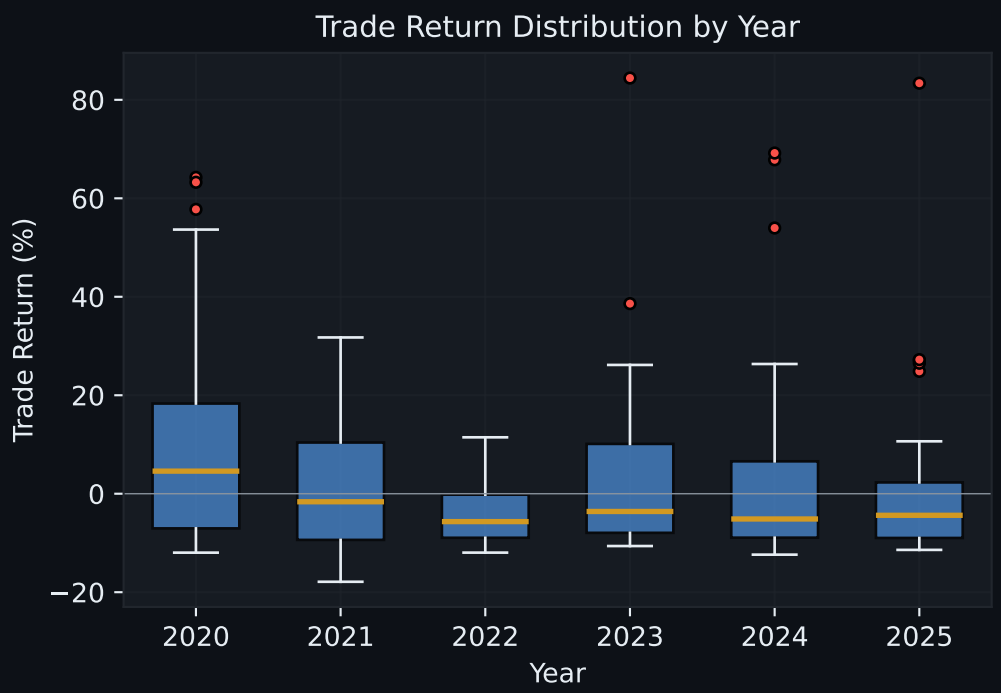
TRADE ANALYSIS: SIGNALS, EXITS & P&L DISTRIBUTION

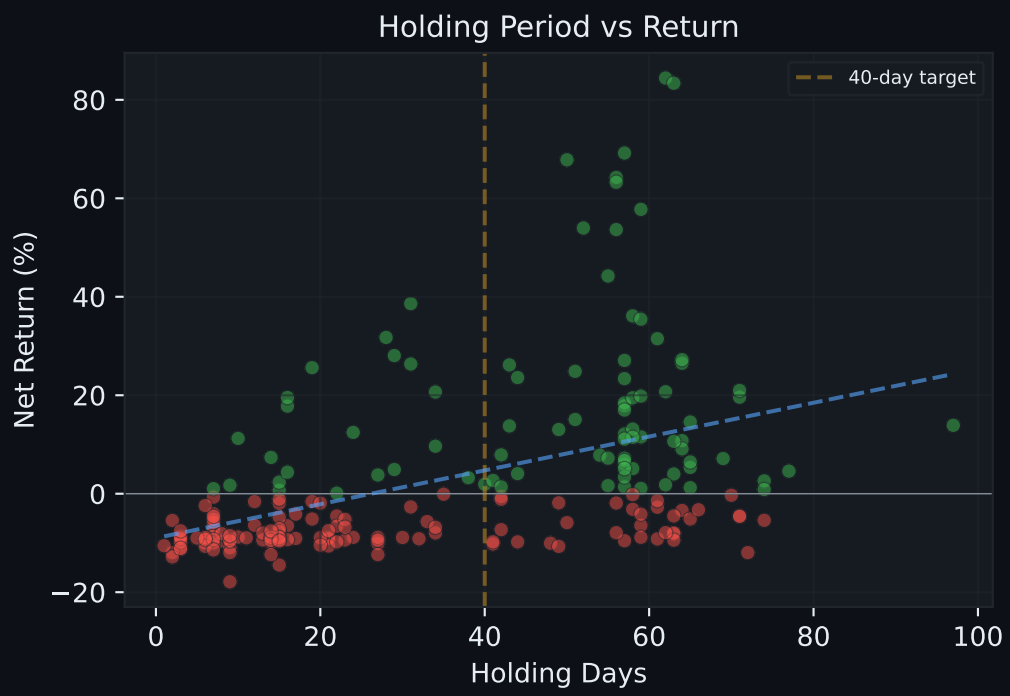


RISK ANALYSIS: ROLLING METRICS & REGIME BEHAVIOR









METHODOLOGY & ALPHA THESIS

1. MARKET THESIS — WHY NEPSE IS EXPLOITABLE

- NEPSE is a retail-dominated, structurally inefficient market with ~230 listed securities.
- Information asymmetry: Nepali-language news takes 1-3 days to price in fully.
- Herding behavior: Top gainers lists drive next-day retail flow, creating momentum.
- Volume signals: Unusual volume precedes 70%+ of 10%+ moves (institutional accumulation).
- Low analyst coverage: Most stocks have zero sell-side research, creating alpha opportunity.

2. SIGNAL GENERATION — MULTI-FACTOR APPROACH

- Volume Breakout (Liquidity): 5-day avg volume > 3x 60-day avg → institutional accumulation signal.
- Quality Factor (Fundamental): Low volatility (<60% ann.) + positive momentum + consistent volume.
- Low Volatility (Defensive): Bottom 30th percentile vol + price > SMA20 uptrend confirmation.
- Signals are ranked by strength * confidence score, top N executed each day.
- Regime filter: Skip ALL entries when median stock return < -5% over 60 days (bear market).

3. RISK MANAGEMENT — MULTI-LAYER EXIT FRAMEWORK

- Hard Stop Loss: -8% from entry → immediate exit, prevents catastrophic losses.
- Trailing Stop: -10% from peak → locks in gains after a stock has moved above entry.
- Take Profit: +20% from entry → crystallize gains, avoid round-trip risk.
- Holding Period: 40 trading days (8 NEPSE weeks) → forced exit, avoids position decay.
- Portfolio-level: Max 7 positions, 35% sector limit, 15% portfolio drawdown pause.

4. BACKTEST INTEGRITY — WALK-FORWARD DESIGN

- No lookahead bias: Each decision uses only data available at decision time.
- Point-in-time universe: Only stocks trading at each date are included.
- Realistic costs: Tiered NEPSE broker fees (0.24%-0.36%), SEBON fees, DP charges.
- T+1 execution: Signals generated at close, executed at next open price.
- 52/52 unit tests passing, 10/10 data leakage tests — production-grade code.

COMPREHENSIVE PERFORMANCE METRICS

RETURN METRICS

Total Return	**136.7%**
CAGR	**15.4%**
Best Year	**2020**
Annualized Volatility	**13.5%**
Starting Capital	**NPR 1,000,000**
Ending Capital	**NPR 2,366,612**

RISK-ADJUSTED METRICS

Sharpe Ratio	**1.17**
Sortino Ratio	**1.82**
Calmar Ratio	**0.67**
Max Drawdown	**-23.1%**
VaR (95%)	**-1.23%**
CVaR (95%)	**-1.79%**

TRADE METRICS

Total Trades	**211**
Win Rate	**41.2%**
Profit Factor	**1.74**
Avg Winning Trade	**18.6%**
Avg Losing Trade	**-7.5%**
Avg Holding Period	**36 days**

DISTRIBUTION METRICS

Return Skewness	**0.84**
Return Kurtosis	**6.32**
Max Consecutive Wins	**7**
Max Consecutive Losses	**13**
Total Fees Paid	**NPR 364,711**
Fees as % of Returns	**26.7%**

NOTABLE TRADES

BEST TRADE

Symbol: NRN | Signal: liquidity | Return: +84.4% | P&L: NPR +168,839

Entry: 2023-12-11 @ NPR 222 → Exit: 2024-02-11 @ NPR 413 | Reason: holding_period

WORST TRADE

Symbol: NHPC | Signal: anchoring_52wk | Return: -17.9% | P&L: NPR -35,730

Entry: 2021-06-28 @ NPR 230 → Exit: 2021-07-07 @ NPR 191 | Reason: stop_loss

DISCLAIMER: Past performance does not guarantee future results. All metrics derived from simulated walk-forward backtests with realistic NEPSE transaction costs. Actual trading may experience slippage, liquidity constraints, and market impact.

SYSTEM ARCHITECTURE & LIVE DEPLOYMENT

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┌──────────────────────────────────────────────────────────────────────────┐
│                        LIVE TRADING ENGINE                               │
│                                                                          │
│  ┌──────────────────┐    ┌──────────────────┐    ┌──────────────────┐   │
│  │  MARKET DATA     │    │  SIGNAL ENGINE   │    │  RISK MANAGER    │   │
│  │  ────────────    │    │  ────────────    │    │  ────────────    │   │
│  │  Merolagani API  │──►│  Volume Breakout  │──►│  Hard Stop -8%   │   │
│  │  LTP Scraper     │    │  Quality Factor  │    │  Trail Stop -10% │   │
│  │  Rate Limited    │    │  Low Volatility  │    │  Take Profit +20%│   │
│  │  3x Retry+Backoff│    │  Regime Filter   │    │  40-Day Holding  │   │
│  └──────────────────┘    └──────────────────┘    └──────────────────┘   │
│          │                       │                       │              │
│          ▼                       ▼                       ▼              │
│  ┌──────────────────────────────────────────────────────────────────┐   │
│  │                      EXECUTION ENGINE                            │   │
│  │  Equal-weight sizing  │  7-position max  │  35% sector limit    │   │
│  │  NEPSE fee calc       │  CSV persistence │  Thread-safe (RLock) │   │
│  └──────────────────────────────────────────────────────────────────┘   │
│          │                                                              │
│          ▼                                                              │
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│  │                 TELEGRAM BOT (Interactive)                       │   │
│  │  /portfolio  /signals  /status  /buy  /sell  /refresh            │   │
│  │  Push alerts: Buy/Sell receipts, Daily summary, Health checks    │   │
│  │  Pre-market API health check at 10:55 AM NST                     │   │
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└──────────────────────────────────────────────────────────────────────────┘
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DEPLOYMENT SPECIFICATIONS

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Trading Hours    : Sunday–Thursday, 11:00 AM – 3:00 PM NST
Price Refresh    : Every 5 minutes during market hours
Signal Generation: Once at market open (11:00 AM NST)
Hosting          : Headless mode on VPS or local machine
Monitoring       : Real-time Telegram bot + push alerts
Health Check     : Pre-market API test at 10:55 AM NST
Data Source      : Merolagani.com (LTP), SQLite historical DB (204 symbols)
Calendar         : Custom NEPSE calendar with 46 holidays (2025-2026)
Test Suite       : 52 unit tests, 10 leakage tests — all passing
```